Replicating xRFM: Accurate, Scalable, and Interpretable Feature Learning Models for Tabular Data +bonus

COMP9417 Group Project

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1 Introduction

Tabular data remains one of the most common data modalities in applied machine learning, appearing in domains such as materials science, healthcare, e-commerce, and human-resource analytics. Despite the rapid progress of deep learning on images and language, GBDTs such as XGBoost and LightGBM remain difficult to outperform on many tabular tasks [Grinsztajn et al., 2022, Holzmüller et al., 2024].

This project replicates and evaluates xRFM, a recent feature learning approach for tabular data based on the average gradient outer product (AGOP) [Beaglehole et al., 2025, Radhakrishnan et al., 2024]. Instead of relying only on fixed input features, xRFM uses gradient-based information from a learned prediction function to estimate directions in the input space that are most relevant to the target. This produces both a predictive model and an interpretable feature-relevance signal. The central question of our replication is therefore whether AGOP-driven feature learning can offer a competitive alternative to strong tabular baselines, especially when compared with XGBoost, LightGBM, and a multilayer perceptron (MLP).

We evaluate all models under a unified pipeline across five public tabular datasets covering regression and binary classification, large and small sample sizes, and mixed feature types (Diamonds, Superconductivity, Online Shoppers, Stroke, HR Attrition). Our main finding is that tree-based GBDTs lead overall, while xRFM shows selective strength on small-sample and fully numerical tasks. Section 2 describes methodology, Section 3 reports results, and Sections 4–5 discuss findings and conclusions.

2 Methodology

2.1 xRFM: Algorithm and Theoretical Background

xRFM is a tabular feature-learning model based on the Average Gradient Outer Product (AGOP) [Beaglehole et al., 2025, Radhakrishnan et al., 2024]. For a trained prediction function $f : \mathbb{R}^d \to \mathbb{R}$, the AGOP is defined as

$$G_f = \mathbb{E}_x\left[\nabla f(x) \nabla f(x)^\top\right] \in \mathbb{R}^{d \times d}. \tag{1}$$

The diagonal entry $[G_f]_{ii} = \mathbb{E}_x[(\partial f/\partial x_i)^2]$ is the expected squared partial derivative with respect to feature i: a large value indicates rapid change in the model output along coordinate i (important feature); a small value indicates near-insensitivity. The full G_f supports two interpretability strategies: its *diagonal* gives coordinate-wise importance scores for each feature, while its *top eigenvectors* identify multivariate directions in input space along which the model varies most, potentially capturing signal invisible to any single coordinate. We use diagonal AGOP in our experiments as it is cheaper to store and sufficient for feature-level rankings.

Compared with alternative methods, AGOP is supervised and model-specific: it reflects the gradient geometry of the fitted function, unlike PCA which only uses input covariance [Pearson, 1901]. Mutual information measures marginal feature–target dependence but not the model's multivariate sensitivity [Kraskov et al., 2004]. Permutation importance is model-agnostic but measures post-hoc performance degradation rather than learned geometry [Fisher et al., 2019]. AGOP is extracted as a by-product of training and directly summarises which directions the learned model relies on.

Training pipeline. xRFM builds a binary tree over the training data. At each node, a base model f is fitted to the local data partition and its AGOP G_f is computed. The leading eigenvector v_1 of G_f defines the split direction; the threshold t^* minimises weighted target

variance along the projection $v_1^\top x$. Each leaf fits a local random Fourier feature kernel machine with the feature metric reweighted by G_f. For inference on a new point x^*, the tree is traversed via $v_k^\top x^* \leq t_k$ at each internal node, and the leaf model produces the final prediction. The dominant training cost is AGOP computation: $O(nd)$ per node for a linear approximation, giving total complexity $O(nd \cdot 2^D)$ for tree depth D [Beaglehole et al., 2025]. This addresses the fundamental limitation of standard kernel machines on tabular data — that a global Euclidean metric treats all features equally — by adaptively reweighting feature space at each tree node.

2.2 Experimental Design

2.2.1 Datasets

Five public tabular datasets (Table 1) are used, spanning regression and binary classification, small and large sample sizes, and mixed feature types [Hamidieh, 2018, Sakar et al., 2019, fedesoriano, 2021].

Table 1: Dataset statistics. n_{train}, n_{val}, and n_{test} denote the sizes of the training, validation, and test splits respectively; d is the number of features after preprocessing.

Dataset	Task	Target	n_{train}	n_{val}	n_{test}	d
Diamonds	Regression	price	32,365	10,789	10,789	26
Superconductivity	Regression	critical_temp	12,757	4,253	4,253	81
Online Shoppers	Classification	Revenue	7,398	2,466	2,466	27
Stroke	Classification	stroke	3,066	1,022	1,022	21
HR Attrition	Classification	Attrition	882	294	294	51

2.2.2 Preprocessing and Splitting

All experiments use seed 42 and a 60/20/20 split. Numerical features are z-score normalised and categorical features are one-hot encoded; missing values are median/constant imputed. The same encoded matrix is used across all four models.

2.2.3 Evaluation Metrics

Regression tasks are evaluated with RMSE; binary classification tasks with accuracy and AUC-ROC. Training and inference time are also recorded.

2.2.4 xRFM Configuration

The xRFM experiments use the Python `xrfm` package through the shared project training pipeline. The model is initialised as `xRFM(use_diag=True, device=DEVICE)`, where `DEVICE` is CUDA if available and CPU otherwise. On GPU, the training script sets the number of fit iterations to 3 and `M_batch_size=128` to control memory cost. Feature importance is extracted by reading the learned AGOP matrix or diagonal values from the trained xRFM model and ranking features by their AGOP-derived scores.

Table 2: xRFM configuration used in the experiments.

Component	Setting
Model	`xRFM`
AGOP mode	Diagonal AGOP, `use_diag=True`
Device	CUDA if available; otherwise CPU
Random seed	42
GPU fit iterations	3
GPU AGOP batch size	128
Data split	60/20/20 train/validation/test
Regression metric	RMSE
Classification metrics	Accuracy and AUC-ROC
Interpretability output	Top features ranked by AGOP diagonal score

2.2.5 XGBoost and LightGBM Configuration

XGBoost and LightGBM are trained through the scikit-learn API using strong default settings from recent tabular benchmarking work [Chen and Guestrin, 2016, Ke et al., 2017, Holzmüller et al., 2024]. Both use 1,000 maximum boosting rounds, learning rate $\eta = 0.05$, row and column subsampling of 0.8, and early stopping with patience 20 on the validation set. XGBoost uses maximum tree depth 6, while LightGBM uses `num_leaves=31`.

2.2.6 MLP Configuration

The MLP baseline uses a two-hidden-layer network with hidden sizes $(128, 64)$ and ReLU activation. It is trained with Adam, learning rate 0.001, batch size 256, maximum 500 iterations, and early stopping with patience 20. Input features are standardised before training. For classification tasks, predicted probabilities are used for AUC-ROC calculation.

3 Results

This section compares xRFM with XGBoost, LightGBM, and MLP in terms of predictive performance, scalability, and interpretability. All models are evaluated under the same preprocessing, train-validation-test split, and metrics defined in Section 2.

3.1 Predictive Performance

Table 3 reports the final held-out test performance across all five datasets. RMSE is used for regression tasks, while accuracy and AUC-ROC are used for binary classification tasks. Bold values indicate the best result for each metric.

Regression tasks. LightGBM achieves the lowest RMSE on both regression datasets. xRFM is close on Superconductivity (9.454 vs. 9.209) but far behind on Diamonds (1496.72 vs. 535.99), indicating that AGOP-based feature learning does not automatically generalise to large mixed-type regression.

Classification tasks. XGBoost and LightGBM achieve the best AUC-ROC on Online Shoppers and Stroke. xRFM's strongest result is HR Attrition, where it achieves the highest accuracy (0.8673), suggesting a modest advantage in small-sample settings. MLP has notably low AUC on Stroke (0.269), reflecting sensitivity to class imbalance.

Table 3: Test-set performance of all models. **Bold** indicates best result per dataset and metric. RMSE is lower-better; Accuracy and AUC are higher-better.

Dataset	Metric	Model			
		xRFM	XGBoost	LightGBM	MLP
Diamonds	RMSE ↓	1496.72	544.68	**535.99**	734.69
Superconductivity	RMSE ↓	9.454	9.302	**9.209**	11.066
Online Shoppers	Accuracy ↑	0.8897	0.9031	**0.9039**	0.8946
	AUC ↑	0.9018	**0.9306**	0.9306	0.9040
Stroke	Accuracy ↑	0.9413	0.9491	0.9501	**0.9511**
	AUC ↑	0.7605	**0.8388**	0.8231	0.2690
HR Attrition	Accuracy ↑	**0.8673**	0.8571	0.8469	0.8639
	AUC ↑	0.7631	0.7552	**0.7886**	0.7512

Table 4 reports training time and per-sample inference time for all models and datasets.

Table 4: Training time (s) and inference time per sample (μs) for all models. Each cell shows Train / Infer.

Dataset	xRFM	XGBoost	LightGBM	MLP
Diamonds	246.5 s / 69.8 μs	1.4 s / 1.9 μs	0.8 s / 1.9 μs	26.8 s / 0.9 μs
Superconductivity	1.6 s / 1.0 μs	3.4 s / 2.1 μs	1.9 s / 7.6 μs	34.3 s / 0.8 μs
Online Shoppers	0.7 s / 0.8 μs	0.2 s / 1.4 μs	0.3 s / 1.6 μs	2.1 s / 0.8 μs
Stroke	0.3 s / 1.1 μs	0.1 s / 1.1 μs	0.1 s / 2.0 μs	0.4 s / 1.0 μs
HR Attrition	0.1 s / 3.2 μs	0.1 s / 3.0 μs	0.1 s / 7.5 μs	0.4 s / 2.2 μs

3.2 Scalability Analysis

To evaluate scalability, we run subsampling experiments on Diamonds and Superconductivity. Table 5 reports RMSE and training time at different training sizes. Figure 1 visualises the Diamonds scalability curves.

Table 5: Scalability experiment: test RMSE and training time (seconds) as a function of training set size. Dash (–) indicates configurations not run.

Dataset	n_{train}	xRFM		XGBoost		LightGBM		MLP	
		RMSE	Time (s)	RMSE	Time (s)	RMSE	Time (s)	RMSE	Time (s)
Diamonds	1,000	1657.0	0.77	846.7	0.80	1003.4	0.13	1081.0	2.27
	5,000	1655.9	11.03	628.9	0.39	622.7	0.11	901.1	10.12
	10,000	1577.6	39.31	587.8	0.49	592.3	0.15	888.9	10.33
	30,000	1487.5	226.44	542.7	0.59	535.3	0.29	743.5	22.23
Supercond.	1,000	14.07	0.08	13.76	0.39	13.91	0.15	15.39	2.52
	5,000	11.30	0.47	10.81	1.00	10.71	0.46	13.05	8.58
	10,000	9.81	1.07	9.51	1.73	9.50	1.10	11.22	28.35
	12,757	9.46	1.56	9.28	1.64	9.21	1.04	–	–

On Diamonds, xRFM's training time grows from 0.77s to 226.44s as n increases from 1,000 to 30,000, while its RMSE only improves from 1657.0 to 1487.5; LightGBM reaches 535.3 in under

one second. On Superconductivity, xRFM scales more favourably, improving from 14.07 to 9.46 RMSE over the same range and remaining within 2.7% of LightGBM at full size. This contrast suggests xRFM is better suited to fully numerical, high-dimensional regression than to large mixed-type datasets with strong threshold effects.

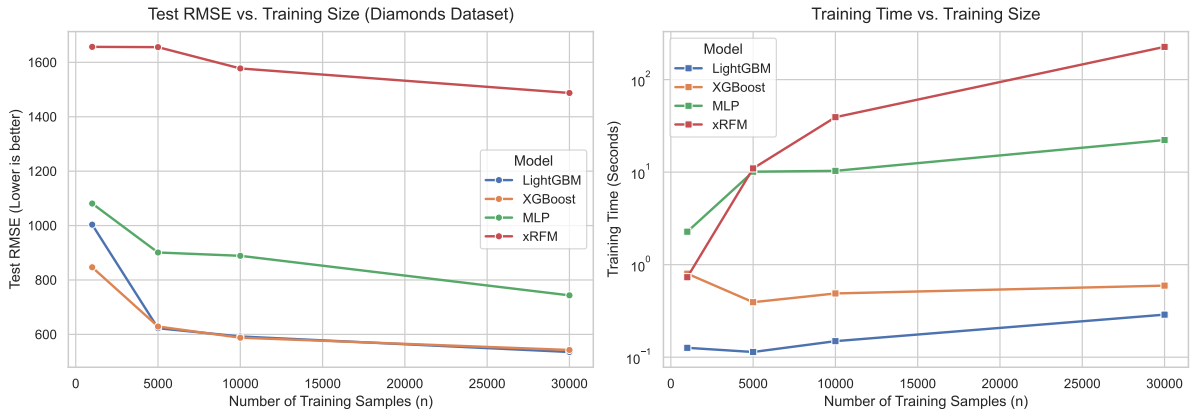

Figure 1: Scalability analysis on the Diamonds dataset, comparing test RMSE (left) and training time (right) across varying training set sizes.

3.3 Interpretability Comparison

Table 6 lists the top-five AGOP-ranked features extracted from each trained xRFM model.

Table 6: xRFM AGOP-diagonal feature importance extracted from the trained models.

Dataset	Top AGOP-ranked features, from 1st to 5th
Diamonds	`y`, `x`, `carat`, `z`, `depth`
Superconductivity	`number_of_elements`, `mean_atomic_mass`, `wtd_mean_atomic_mass`, `gmean_atomic_mass`, `wtd_gmean_atomic_mass`
Online Shoppers	`PageValues`, `ExitRates`, `ProductRelated`, `ProductRelated_Duration`, `BounceRates`
Stroke	`avg_glucose_level`, `age`, `bmi`, `work_type_Self-employed`, `smoking_status_smokes`
HR Attrition	`OverTime_No`, `OverTime_Yes`, `BusinessTravel_Travel_Frequently`, `TotalWorkingYears`, `BusinessTravel_Non-Travel`

The AGOP rankings are broadly interpretable: physical size variables for Diamonds, `PageValues` for Online Shoppers, glucose/age/BMI for Stroke, and overtime indicators for HR Attrition all align with domain knowledge.

Table 7 compares AGOP diagonal importance with PCA, mutual information, and permutation importance. These methods answer different questions. AGOP measures which features the learned xRFM function is sensitive to; PCA measures high-variance input directions; mutual information measures feature-target dependence; and permutation importance measures how much test performance drops when a feature is shuffled.

Table 7: Conceptual comparison of interpretability methods for tabular features.

Method	Supervised?	Model-specific?	Main interpretation
AGOP diagonal	Yes	Yes	Sensitivity of xRFM to each feature
PCA loadings	No	No	Directions of maximum input variance
Mutual information	Yes	No	Feature-target dependence
Permutation importance	Yes	Model-agnostic	Performance drop after feature shuffle

Notably, meaningful AGOP rankings do not imply strong predictive performance: on Diamonds, AGOP identifies sensible size-related features, yet xRFM's RMSE is nearly three times higher than LightGBM's. This suggests that boosted trees capture axis-aligned thresholds more efficiently than AGOP reweighting, and that interpretability and accuracy can be decoupled.

4 Discussion

4.1 xRFM: When Feature Learning Helps and When It Does Not

xRFM's strongest result is HR Attrition ($n = 882$), where its structured feature-learning prior may help in low-data settings. However, on Diamonds it obtains RMSE 1496.72 versus 535.99 for LightGBM — nearly three times worse. Diamonds contains strong axis-aligned categorical effects (cut, colour, clarity) that boosted trees capture efficiently through split-based interactions [Chen and Guestrin, 2016, Ke et al., 2017], whereas AGOP gradient reweighting is less effective here. The scalability experiment confirms that adding more Diamonds data does not close this gap. In contrast, on the fully numerical Superconductivity dataset xRFM approaches the tree baselines, suggesting it is better suited to continuous high-dimensional inputs. Future work should target categorical handling, more efficient AGOP approximation, and dataset-specific tuning.

4.2 XGBoost and LightGBM: Performance and Efficiency

XGBoost and LightGBM deliver competitive or superior performance across all five datasets while remaining the fastest models to train. On Diamonds, both GBDTs outperform xRFM by a large margin in RMSE (544.68 and 535.99 versus 1496.72), consistent with the strong axis-aligned inductive bias of tree-based splits [Grinsztajn et al., 2022]. On HR Attrition ($n = 882$, $d = 51$), xRFM marginally surpasses XGBoost in accuracy (0.8673 versus 0.8571), suggesting that kernel-style feature learning offers a modest advantage only in a restricted small-data regime. LightGBM is the fastest model at every large Diamonds scale, with near-linear growth in training time (Table 5).

4.3 MLP: Performance and Scalability Limitations

MLP underperforms both GBDTs on regression (RMSE 743.5 vs. 535.3 on Diamonds at $n =$ 30,000) and shows unstable classification results, most notably AUC 0.269 on Stroke. This reflects sensitivity to class imbalance and the limited alignment between global gradient-based optimisation and the piecewise structure of tabular features [Holzmüller et al., 2024].

5 Conclusion

We replicated xRFM and compared it with XGBoost, LightGBM, and MLP across five tabular datasets under a unified experimental pipeline. Tree-based GBDTs remain the strongest general-purpose baselines: LightGBM and XGBoost lead on accuracy and training speed across nearly all settings. xRFM shows selective promise — competitive on Superconductivity and best

accuracy on HR Attrition — but underperforms substantially on Diamonds and scales poorly there, indicating that AGOP feature learning is not universally superior to tree splitting on large mixed-type data. AGOP-diagonal rankings are broadly interpretable and align with domain knowledge, though meaningful rankings do not guarantee predictive strength. Future work should address categorical variable handling, AGOP approximation efficiency, and hyperparameter tuning for diverse tabular settings.

A AGOP-Based Splitting Criterion: Scratch Implementation

The xRFM tree selects the split direction at each node as the leading eigenvector v_1 of the AGOP $G_f = \mathbb{E}_x[\nabla f(x) \nabla f(x)^\top]$ [Beaglehole et al., 2025]. We implement this criterion from scratch using Ridge regression ($f(x) = w^\top x + b$, $\alpha = 10^{-3}$). For a linear model the gradient is constant so the AGOP simplifies to $G_f = ww^\top$, giving the closed-form diagonal $[G_f]_{ii} = w_i^2$ and split direction $v_1 = w/\|w\|$. The optimal threshold t^* is found by scanning 30 candidate quantiles of $v_1^\top X$ and picking the one that minimises weighted target variance. The full implementation is in `bonus/agop_split.py`.

Verification. We use a synthetic dataset ($n = 300$, $d = 10$) with $y = 3x_0 - 2x_1 + \varepsilon$, so the ground-truth signal features are $\{x_0, x_1\}$. We compare the scratch AGOP diagonal w_i^2 against a numerical AGOP diagonal computed by finite-difference gradients of the fitted xRFM predictor (black-box use only).

Table 8: AGOP diagonal on the synthetic dataset. Noise features x_2–x_9 score ≤ 0.03 (scratch) and ≤ 0.30 (xRFM).

Feature	Scratch w_i^2	xRFM numerical
x_0 (coef. +3)	9.23	13.03
x_1 (coef. −2)	3.93	5.28
x_2–x_9 (noise)	≤ 0.03	≤ 0.30
Top-2 identified	$\{x_0, x_1\}$	$\{x_0, x_1\}$
Cosine similarity	**0.9985**	

Both methods identify $\{x_0, x_1\}$ correctly (overlap 2/2, cosine similarity 0.9985), confirming that the scratch criterion recovers the same split direction as the xRFM library.

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